Lists and loops

Programming · Level 3 · Lesson 2



In this lesson you will

- Index a list from 0 and from the end with -1.
- Count, total and find the largest in a loop.
- Use a list as a stack or a queue.

Key ideas

- temps[0] is first; temps[-1] is last.
- Largest: start with the first item, not 0.
- Stack: pop(). Queue: pop(0).

Activity: The human list

1. Warm-up (3 min): Five classmates hold number cards in a row.
2. Index (7 min): Call out temps[0], temps[-1], temps[1:3].
3. Loop (7 min): Find the largest, one card at a time.
4. Stack (3 min): Make a pile of books: push and pop.

Use made-up readings, and include a negative one: it is the case that breaks “start at 0”.

Let’s discuss

1. Why does largest start with the first item, not 0?
2. Where do you meet stacks and queues in real life?
3. What happens when you ask for an index that doesn’t exist?

Exit ticket

- Write a loop that counts the even numbers in a list.